s5_sector_neutral

equity strategy | family: equity-neutralized | status: validated | derived from `s4_tilted_size_neutral`

What this strategy does

Trim 0.15, linear tilt, size x FF12-sector neutral, 1.5x leverage cap

Why it should work. Same drift, with sector and size risk diversified away so only the earnings signal is left.

Design notes (from the strategy's source file).

Strategy 5 -- Strategy 4 + FF12-sector neutrality.

A diagnostic found Strategy 4's book putting up to 29% of gross weight in one FF12 sector in some quarters (8.3% if spread evenly): an unpriced bet unrelated to earnings drift. Neutralizing each (side, size, sector) cell to an equal budget removes it. Best Sharpe / shallowest drawdown of the hand-built strategies at its conservative 1.5x leverage cap.

parameter	value
base_unit_fraction	0.0015
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

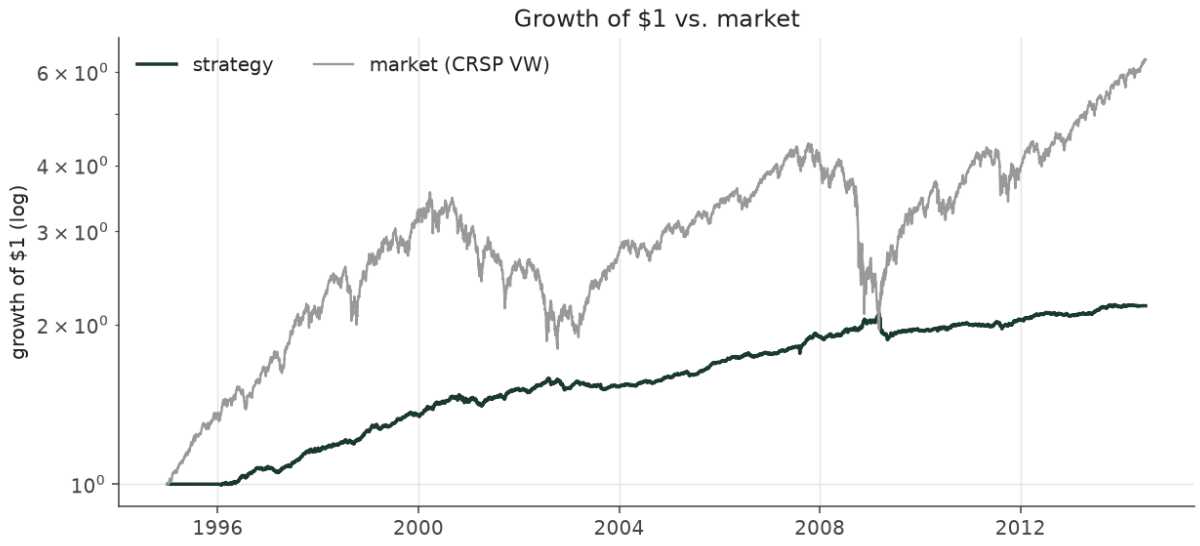
metric	s5 sector neutral	market (CRSP VW)
Annualized return	4.06%	9.94%
Annualized volatility	3.02%	19.35%
Sharpe (NAV-pct)	1.33	0.59
Sortino	1.88	
Max drawdown	-10.8%	-55.5%
Calmar	0.37	
Final NAV ($10M start)	$21,709,852	
Worst day	-3.95%	
% positive days	52.3%	
Sharpe 1996-2001	2.19	
Sharpe 2002-2007	1.34	
Sharpe 2008-2013	0.66	

Correlation of daily returns with the market: **0.03**.

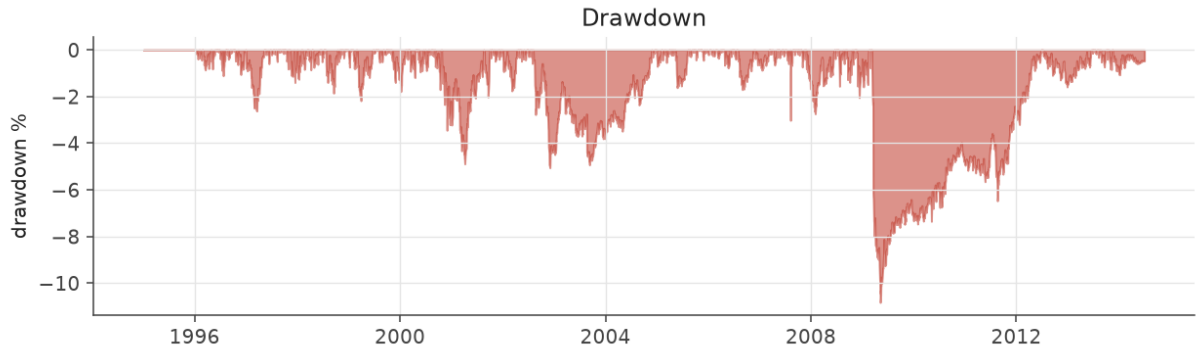
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 1.33

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.0406

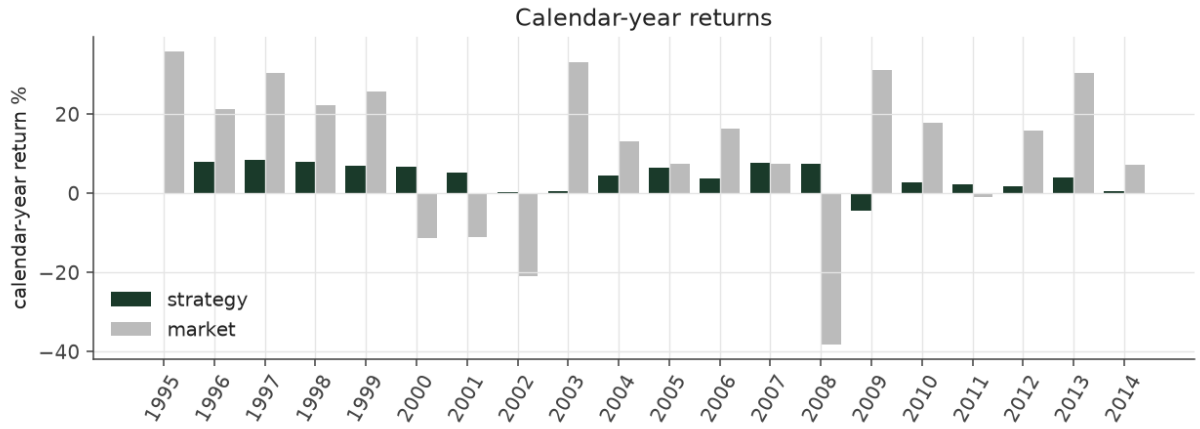
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.375



NAV (log scale)



Drawdown from running peak

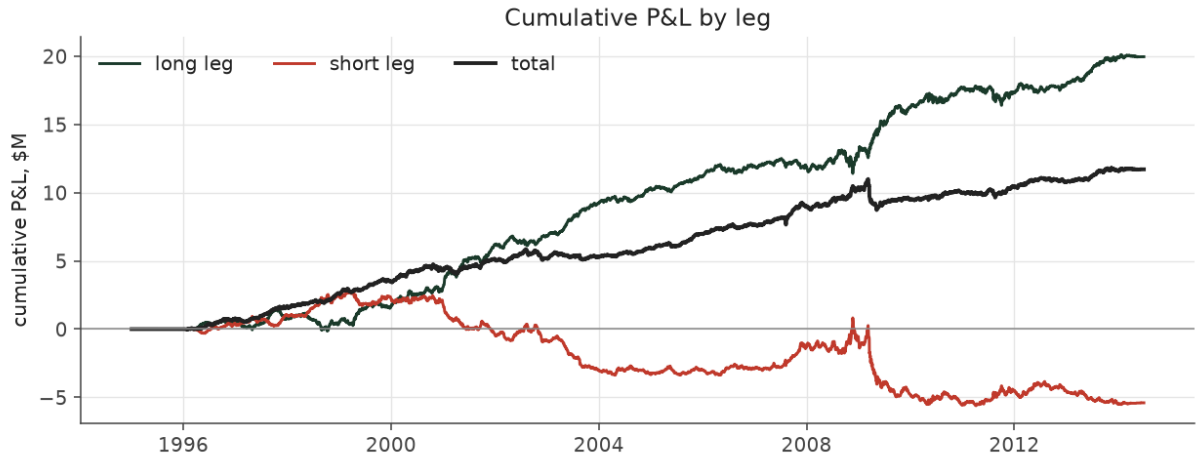


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	19.96	-5.40
P&L after costs ($M)	18.59	-6.88
of which common (average-event) component ($M)	10.40	-10.71
of which excess / PEAD component ($M)	9.56	5.31
excess minus costs ($M)	8.18	3.83
gross return per $ traded (bp)	304.72	-80.72
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.32	-0.31

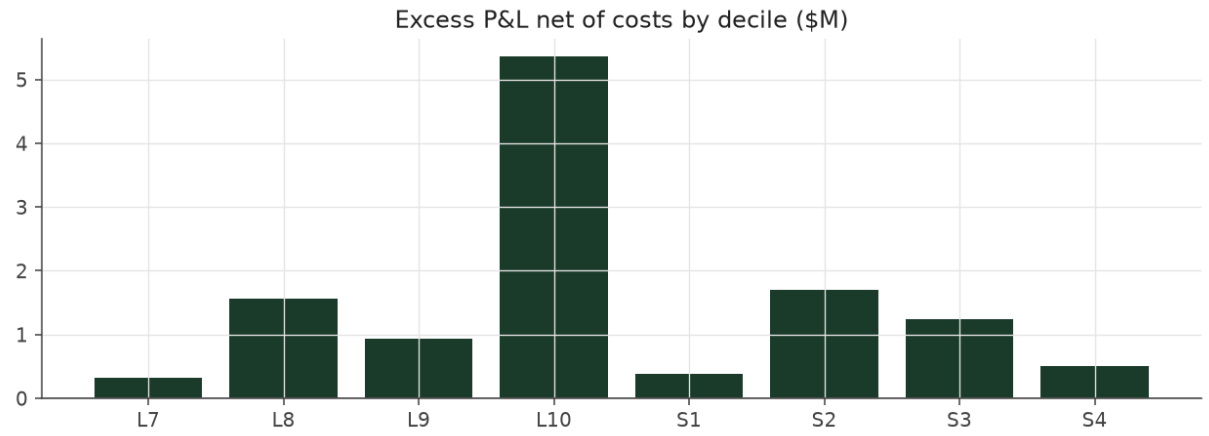
Daily P&L correlation between the two legs: **-0.79**.

• **The short leg loses money in absolute terms** (-5.4M before costs) while the long leg earns +20.0M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 10.4M / 10.7M. Stripping that common component, the surprise signal itself earned +9.6M on the long side and +5.3M on the short side.

• The two legs' daily P&L are -0.79 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.32 long / -0.31 short vs 1.33 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	56.48	1.39	0.44	0.12	0.32	2.7%	56.82
long	8	16,588	151.86	4.08	1.88	0.32	1.56	13.0%	102.79
long	9	17,747	202.60	3.85	1.36	0.42	0.94	7.8%	46.36
long	10	17,174	244.17	9.27	5.88	0.52	5.36	44.6%	219.67
short	1	17,169	247.53	-3.33	0.96	0.57	0.39	3.2%	15.64
short	2	17,281	208.23	-1.82	2.15	0.45	1.70	14.1%	81.65
short	3	15,970	152.89	-1.24	1.57	0.33	1.25	10.4%	81.44
short	4	9,171	60.77	-0.48	0.62	0.12	0.50	4.2%	82.45



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	164.15	4.46	5.45	0.66	4.79	39.9%	291.79
2	22,885	262.98	2.30	3.15	0.76	2.40	19.9%	91.10
3	25,501	294.17	2.65	3.24	0.63	2.61	21.7%	88.78
4	27,583	301.86	1.36	1.81	0.47	1.34	11.1%	44.33
5	28,947	301.38	0.94	1.22	0.33	0.88	7.3%	29.31

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	130.20	3.85	4.71	0.65	4.06	33.8%	312.06
2	23,084	287.52	2.88	4.06	0.86	3.20	26.6%	111.25
3	25,851	308.05	2.25	3.00	0.62	2.39	19.9%	77.54
4	27,665	300.81	1.87	2.08	0.42	1.66	13.8%	55.08

5	28,874	297.88	0.86	1.01	0.30	0.71	5.9%	23.82

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	481.23	0.05	2.79	0.88	1.91	15.9%	39.62
41-60d	25,652	300.47	1.88	2.18	0.66	1.52	12.6%	50.59
61-90d	35,324	395.08	2.86	3.57	0.90	2.68	22.3%	67.74
>90d	12,298	147.76	6.92	6.33	0.41	5.92	49.2%	400.37

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	328.62	4.94	5.76	0.69	5.07	42.2%	154.33
2002-07	39,956	438.83	4.14	5.12	0.92	4.19	34.9%	95.58
2008-13	45,278	557.10	2.64	3.99	1.24	2.75	22.9%	49.41

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	124.11	1.01	1.28	0.27	1.01	8.4%	81.08
Chems	2,590	92.50	0.28	0.50	0.18	0.32	2.7%	34.65
Durbl	2,843	101.51	1.37	1.60	0.21	1.39	11.6%	137.24
Enrgy	6,194	109.07	1.62	1.88	0.22	1.65	13.7%	151.51
Hlth	11,696	121.82	1.26	1.55	0.27	1.28	10.7%	105.12
Manuf	12,034	116.86	0.99	1.31	0.26	1.05	8.8%	90.15
Money	17,583	114.16	1.27	1.58	0.27	1.30	10.8%	114.23
NoDur	5,081	108.74	0.87	1.08	0.23	0.85	7.1%	78.58
Other	18,377	121.67	1.17	1.49	0.28	1.21	10.0%	99.03
Shops	11,984	120.09	1.49	1.76	0.26	1.50	12.5%	124.88
Telcm	3,981	108.87	0.26	0.49	0.22	0.28	2.3%	25.25
Utils	4,672	85.15	0.11	0.34	0.17	0.17	1.4%	20.45

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	411.62	2.13	3.39	0.80	2.60	21.6%	63.04
low vol	40,091	462.76	4.81	5.96	1.10	4.87	40.5%	105.17
mid vol	37,903	425.63	4.61	5.28	0.90	4.38	36.4%	102.89

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	4.06%	0.00%
0.5%	0.7	3.89%	0.17%
1.0%	1.4	3.71%	0.35%
2.0%	2.8	3.34%	0.72%
5.0%	6.9	2.04%	2.02%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.